

Game Theory - Problem Set 2

Q1 - Matrix Game Analysis

Consider the following strategic game:

P1 \ P2	A	B	C	D
x	8, 5	10, 4	6, 6	8, 6
y	4, 5	10, 8	10, 10	6, 9
z	4, 5	10, 9	7, 4	7, 10

(a)

For all players, find all pure strategies that are **strictly dominated** by other strategies (pure or mixed).

Definition. A pure strategy is strictly dominated if another strategy (pure or mixed) yields strictly greater payoff for all opponent strategy profiles.

We will begin by comparing the pure strategies, for the mixed ones are a bit of a pain.

Player 1.

By looking at our strategy matrix (more of a table to be honest, but also, what is a table of values if not a matrix?). We can see that not one strategy is strictly dominated by another for player 1.

Player 2.

Comparing A versus D.

(5,5) versus (6,9,10) shows D strictly better in all cases so A strictly dominated by D.

Comparing B versus D.

(4,8,9) versus (6,9,10) shows D strictly better in all cases so B strictly dominated by D.

Now we're not free yet, as we must check the mixed strategies that might form here. We could, of course, individually check all of the possible combinations, but that seems like a massive, and unnecessary amount of effort wasted when a more simple alternative exists.

We take a good look at our strategy matrix, we take the mix (suppose that it is x and y) and when comparing it against its opponent (z), we can look at the highest option given to us by the mix (x and y). If even that isn't enough to overcome the value of the opponent (z) then there is not even a point in continuing.

Thus, we will now compare the mixed strategies.

Player 1

Compare z versus mixed strategy of x and y with probability p.

Mixed strategy payoffs versus each P2 strategy:

vs A: $4 + 4p$

vs B: 10

vs C: $10 - 4p$

vs D: $6 + 2p$

Compare with z's payoffs (4, 10, 7, 7):

$4 + 4p > 4 \Rightarrow p > 0$

$10 = 10$ shows equal for all p

$10 - 4p > 7 \Rightarrow p < 0.75$

$6 + 2p > 7 \Rightarrow p > 0.5$

Since we cannot get strictly greater payoff versus B, z is not strictly dominated

No other P1 pure strategy is strictly dominated.

Player 2 Analysis:

We already have two strictly dominated strategies, and taking a single at C shows that there's nothing to be done.

Thus, P1 - None, P2 - A, B

(b)

For all players, find all pure strategies that are **weakly dominated** (but not strictly) by other strategies (pure or mixed).

Definition. A pure strategy is weakly dominated if another strategy yields at least as good payoff for all opponent profiles and strictly better for at least one.

Player 1:

z versus mixed strategy (0.6x, 0.4y):

vs A: $6.4 > 4$

vs B: $10 = 10$

vs C: $7.6 > 7$

vs D: $7.2 > 7$

Thus z is weakly dominated by this mixed strategy

Player 2:

After removing strictly dominated strategies A and B

C versus D: versus y gives $10 > 9$ showing C better here so no dominance

D versus C: versus z gives $10 > 4$ showing D better here so no dominance

Thus, P1: z, P2: None

(c)

For all players, find all pure strategies that are **weakly dominated** (but not strictly) by other strategies (pure or mixed).

Definition. Iterated Elimination of Strictly Dominated Strategies removes strategies that are strictly dominated in each iteration.

Initial: P1: {x,y,z}, P2: {A,B,C,D}

Remove P2: A, B (strictly dominated by D)

Remaining: P1: {x,y,z}, P2: {C,D}

Reduced matrix.

P1 \ P2	C	D
x	6, 6	8, 6
y	10, 10	6, 9
z	7, 4	7, 10

No further strictly dominated strategies can be removed

Thus, IESDS: P1: {x,y,z}, P2: {C,D}

(d)

For all players, find all pure strategies that are **weakly dominated** (but not strictly) by other strategies (pure or mixed).

Definition. Iterated Elimination of Weakly Dominated Strategies removes strategies that are weakly dominated.

Remove P2: A, B (strictly dominated)

Remove P1: z (weakly dominated by mixed x,y)

Reduced matrix.

P1 \ P2	C	D
x	6, 6	8, 6
y	10, 10	6, 9

Thus, IEWDS. P1: {x,y}, P2: {C,D}

(e)

Find all pure NE (if any).

Definition. A Nash Equilibrium occurs when each player's strategy is a best response to the other players' strategies

(x,D):

P1: 8; if switching to y:6 worse, if switching to z:7 worse so no deviation

P2: 6; if switching to C:6 same, if switching to A:5 worse, if switching to B:4 worse so no profitable deviation

Thus (x,D) is a Nash Equilibrium.

(y,C):

P1: 10; if switching to x:6 worse, if switching to z:7 worse so no deviation

P2: 10; if switching to D:9 worse so no deviation

Thus (y,C) is a Nash Equilibrium.

Thus, Pure Nash Equilibria: (x,D), (y,C)

△

Q2 Bidding Game

There are three players and a single indivisible object. Each player i has a value v_i for the object. All players know the values of other players. Values are as follows:

$$v_1 = 7, \quad v_2 = 4, \quad v_3 = 1$$

Each player simultaneously and independently submits bid $b_i \in \{0, 1, 2, \dots\}$ (i.e., bids must be integers) for the object. The winner of the object always has to pay the amount of its own bid while losers pay nothing, i.e., player i gets $v_i - b_i$ when she wins the object, and 0 otherwise. The exact rules for determining the winner of the object will be specified later.

(a)

Suppose that the game rules for determining the winner are fair and symmetric. In other words, if all players bid the same amount, the object is awarded randomly to one of them with fair odds. For each player, identify all weakly dominated strategies in such games.

We claim that for any player i , any bid $b_i > v_i$ is weakly dominated by bidding v_i .

The reason for this is that for any opponents' bids, raising your own bid above your current value can only make you either:

- (i) win and get $v_i - b_i \leq 0$ (worse than losing and getting 0)
- (ii) not change the outcome.

Bidding v_i guarantees not to produce a payoff strictly less than bidding b_i , and in some cases strictly better. So any bid strictly above one's own valuation is weakly dominated by bidding exactly your valuation.

Thus, $\forall i$:

All bids $b > v_i$ are weakly dominated (by $b = v_i$), and no bids $b \leq v_i$ are weakly dominated.

Player 1 ($v_1 = 7$):

Bidding 7 vs bidding 6:

If win with bid 7: payoff $7 - 7 = 0$

If win with bid 6: payoff $7 - 6 = 1$

No scenario where bid 7 gives strictly higher payoff. Thus bid 7 weakly dominated by bid 6. Similarly, bid 6 weakly dominated by bid 5, and onwards. This continues until bid 1, for bid 1 is not weakly dominated by bid 0 because if opponents bid 0, bid 1 wins with payoff $7 - 1 = 6$ while bid 0 gives expected payoff $\frac{1}{3} \times 7 \approx 2.33$

Player 2 ($v_2 = 4$):

Bids ≥ 4 weakly dominated by lower bids, the maximum undominated bid is 3.

Player 3 ($v_3 = 1$):

Bids ≥ 1 weakly dominated by lower bids. Thus the only undominated bid is 0.

Thus, weakly dominated strategies for an individual player are.

Player 1. $\{7, 8, 9, \dots\}$

Player 2. $\{4, 5, 6, \dots\}$

Player 3. $\{1, 2, 3, \dots\}$

(b)

Suppose that the winner of the object is the player with the **highest bid**. Ties (if any) are broken randomly using a fair randomizing device.

Find all pure-strategy Nash equilibria of this game in which players **do not use** weakly dominated strategies identified in part (a).

Values: $v_1 = 7, v_2 = 4, v_3 = 1$

After eliminating weakly dominated strategies from part (a), we consider:

- $b_1 \in [0, 7)$
- $b_2 \in [0, 4)$
- $b_3 = 0$

Player 1 will not bid less than 4, because:

If $b_1 < 4$, player 2 can have a bigger bid and Player 1 would get 0, which is worse than winning with a bid of 4.

At the same time, b_1 will not be bigger than 4 because:

For $b_1 > 4$, payoff $7 - b_1 < 3$ with $b_1 = 4$, player 1 will win

Player 2 is indifferent for $b_2 = 0, 1, 2, 3$ (in all cases gets 0)

Player 3 must bid $b_3 = 0$

Thus, the pure-strategy Nash equilibria are:

$$NE : (4, b_2, 0) \quad \text{where} \quad b_2 \in \{0, 1, 2, 3\}$$

1. $(4, 0, 0)$
2. $(4, 1, 0)$
3. $(4, 2, 0)$
4. $(4, 3, 0)$

In all these equilibria, Player 1 (with the highest valuation $v_1 = 7$) wins the object. This outcome is **socially efficient** because it maximizes social welfare as the object goes to the player who values it most.

△

(c)

Now suppose that the winner of the object is the player with the **second-highest bid**. Ties (if any) are broken randomly using a fair randomizing device. *For example, if $b_1 = 3, b_2 = 1, b_3 = 2$, then the winner is Player 3. If $b_1 = 1, b_2 = 1, b_3 = 2$, then the winner is either Player 1 (50% chance) or Player 2 (50% chance).*

Find all pure-strategy Nash equilibria of this game in which players **do not use** weakly dominated strategies identified in part (a). For each Nash equilibrium you find, state whether the outcome is socially efficient (maximizes the social welfare).

Thanks to (a) we can already exclude a good chunk of potential bids.

Then, we can easily notice that there are only two realistic options. $(0, 0, 0)$ and $(1, 1, 1)$. The reason for this is that for any player that we pick, they will have to worry about the bid that is weaker to it. The only player that does not have this problem is the third one, the options of bid for whom are 0 or 1.

△

Q3 Let's Split 10K

Players 1 and 2 are bargaining over splitting ten (10) thousand roubles. Both players simultaneously name shares they would like to get, s_1 and s_2 in **whole thousands** of roubles (i.e., both s_1 and s_2 must be a number from $\{0, 1, 2, \dots, 10\}$ set). If $s_1 + s_2 \leq 10$, then players receive the shares they named; if $s_1 + s_2 > 10$, then both players receive zero.

(a)

Formulate a strategic form game that corresponds to this interaction

Strategy sets: $S_1 = \{0, 1, \dots, 10\}$, $S_2 = \{0, 1, \dots, 10\}$ for Players 1 and 2.

Payoffs:

$$u_1(s_1, s_2) = \begin{cases} s_1 & s_1 + s_2 \leq 10 \\ 0 & s_1 + s_2 > 10 \end{cases}$$

$$u_2(s_1, s_2) = \begin{cases} s_2 & s_1 + s_2 \leq 10 \\ 0 & s_1 + s_2 > 10 \end{cases}$$

(b)

Find best response functions for both players and PLOT them on the same graph.

Player 1.

Given s_2 , Player 1 wants to maximize u_1 :

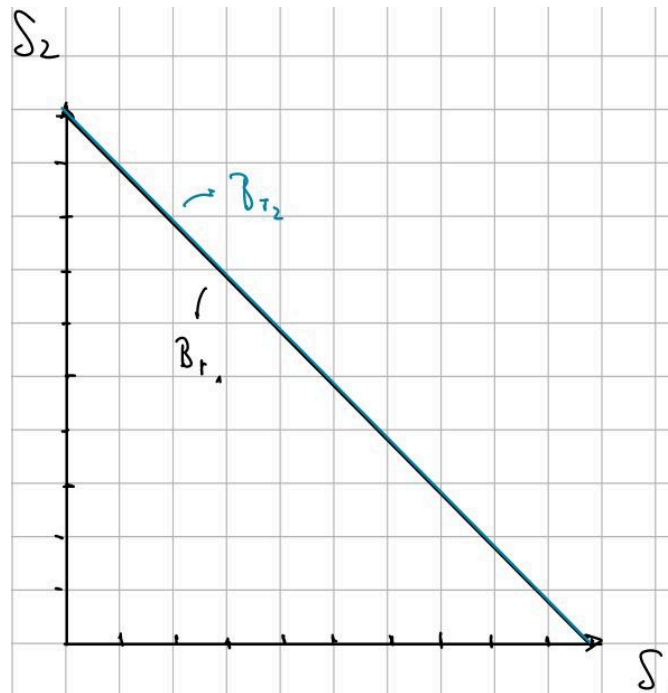
If $s_1 + s_2 \leq 10$, then $u_1 = s_1$, so best is largest s_1 satisfying $s_1 \leq 10 - s_2$

If $s_1 + s_2 > 10$, then $u_1 = 0$

Thus, $BR_1(s_2) = 10 - s_2$

Player 2.

Likewise, $BR_2(s_1) = 10 - s_1$



(c)

Find all pure strategy Nash equilibria.

Nash equilibrium requires:

$$s_1 \in BR_1(s_2) \rightarrow s_1 = 10 - s_2 \quad \text{and} \quad s_2 \in BR_2(s_1) \rightarrow s_2 = 10 - s_1$$

$$\text{Substituting: } s_1 = 10 - (10 - s_1) \implies s_1 = s_1$$

All pairs (s_1, s_2) satisfying $s_1 + s_2 = 10$ would be Nash equilibria.

Thus,

$$(s_1, s_2) \quad s_1 + s_2 = 10, \quad s_1, s_2 \in \{0, 1, \dots, 10\}$$

As a result, are 11 pure Nash equilibria, $(0, 10), (1, 9), (2, 8), \dots, (10, 0)$.

Q4 “Your strength can be your weakness”

A three-person committee has to choose a winner for a national art prize. After some debate, there are three candidates still under consideration: a woman who draws antelope in urban settings, a man who makes rectangular lead boxes, and a woman who sculpts charcoal. Let's call these candidates a, b and c; and call the committee members 1, 2 and 3. The preferences of the committee members are as follows: member 1 prefers a to b and b to c; member 2 prefers c to a and a to b; and member 3 prefers b to c and c to a. Each committee member casts a single vote for one of the candidates. The candidate who received the most votes wins. If the vote is tied (nobody received the majority), the winner is the candidate for whom member 1 voted. Thus, the committee member 1 has an advantage.

Our first step would be assignment preferences.

The preferences are:

- **Voter 1:** $a \succ b \succ c$ (payoff: $a=3, b=2, c=1$)
- **Voter 2:** $c \succ a \succ b$ (payoff: $c=3, a=2, b=1$)
- **Voter 3:** $b \succ c \succ a$ (payoff: $b=3, c=2, a=1$)

Rule: Candidate with most votes wins. Tiebreaker: Voter 1's vote decides.

(a)

Construct best response for each player.

Voter 1: $BR_1(s_2, s_3)$

s_2	s_3	a	b	c	BR_1	Reason
a	a	a	a	a	{a,b,c}	Indifferent - a wins regardless
a	b	a	b	a	{a}	$a=3, b=2, c=1$
a	c	a	a	c	{a}	a-b tie, V1 breaks for a
b	a	a	b	a	{a}	$a=3, b=2, c=1$
b	b	b	b	b	{a,b,c}	Indifferent - b wins regardless
b	c	a	b	c	{a}	$a=3, b=2, c=1$
c	a	a	a	c	{a}	a-c tie, V1 breaks for a
c	b	a	b	c	{a}	$a=3, b=2, c=1$
c	c	c	c	c	{a,b,c}	Indifferent - c wins regardless

Thus:

$s_2 \setminus s_3$	a	b	c
a	{a,b,c}	{a}	{a}
b	{a}	{a,b,c}	{a}
c	{a}	{a}	{a,b,c}

Voter 2: $BR_2(s_1, s_3)$

s_1	s_3	a	b	c	BR_2	Reason
a	a	a	a	a	{a,b,c}	Indifferent
a	b	a	b	a	{a,c}	a=2, b=1, c=2
a	c	a	a	c	{c}	a=2, b=1, c=3
b	a	a	b	a	{a}	a-b tie, V1 breaks for a
b	b	b	b	b	{a,b,c}	Indifferent
b	c	a	b	c	{c}	a=2, b=1, c=3
c	a	a	c	c	{b,c}	b-c tie, V1 breaks for b
c	b	a	b	c	{a,c}	a=2, b=1, c=3
c	c	c	c	c	{a,b,c}	Indifferent

Thus:

$s_1 \setminus s_3$	a	b	c
a	{a,b,c}	{a,c}	{c}
b	{a}	{a,b,c}	{c}
c	{b,c}	{a,c}	{a,b,c}

eVoter 3: $BR_3(s_1, s_2)$

s_1	s_2	a	b	c	BR_3	Reason
a	a	a	a	a	{a,b,c}	Indifferent
a	b	a	b	a	{b}	a=1, b=3, c=1
a	c	a	a	c	{c}	a=1, b=1, c=2
b	a	a	b	a	{b,c}	b=3, c=2, a=1
b	b	b	b	b	{a,b,c}	Indifferent
b	c	a	b	c	{a,b}	b=3, c=2, a=1
c	a	a	c	c	{b,c}	b=3, c=2, a=1
c	b	a	b	c	{b}	b=3, c=2, a=1
c	c	c	c	c	{a,b,c}	Indifferent

Thus:

$s_1 \setminus s_2$	a	b	c
a	{a,b,c}	{b}	{c}
b	{b,c}	{a,b,c}	{a,b}

$s_1 \setminus s_2$	a	b	c
c	{b,c}	{b}	{a,b,c}

(b)

Consider this voting game. Each voter has three strategies: a, b, or c. For each voter, which strategies are weakly or strictly dominated? Hint: Be especially careful in the case of voter 1. To answer this question, you do not need to know the exact payoffs: any payoffs will do provided that they are consistent with the preference orders given above. To answer this question, you do not have to write out matrices.

Solution:

A strategy σ is **strictly dominated** by τ (pure or mixed) if τ gives strictly higher payoff against every possible strategy profile of the others. It is **weakly dominated** (but not strictly) if τ gives at least as high payoff everywhere and strictly higher in at least one case.

Voter 1

b is weakly dominated by a: $a \geq b$ in all cases, $>$ in (a,b), (b,c), (c,b)

c is weakly dominated by a: $a \geq c$ in all cases, $>$ in (a,b), (a,c), (b,c), (c,a), (c,b)

Voter 2

b is weakly dominated by c: $c \geq b$ in all cases, $>$ in (a,c), (b,c), (c,a), (c,b)

Voter 3

a is weakly dominated by b: $b \geq a$ in all cases, $>$ in (a,b), (a,c), (b,a), (c,b)

Thus, no strategies are strictly dominated.

(c)

Now consider the reduced game in which all weakly and strictly dominated strategies have been removed. For each voter, which strategies are now weakly or strictly dominated? What is the predicted outcome of the vote? Compare this outcome to voter 1's preferences and comment.

Step 1: We will remove weakly dominated strategies

Voter 1: Remove b, $c \rightarrow \{a\}$

Voter 2: Remove b $\rightarrow \{a, c\}$

Voter 3: Remove a $\rightarrow \{b, c\}$

Step 2: Analyze reduced game

Voter 2 in reduced game ($s_1 = a$):

- If $s_3 = b$: $a=2, c=2 \rightarrow \{a,c\}$
 - If $s_3 = c$: $a=2, c=3 \rightarrow \{c\}$
- Thus, a is weakly dominated by c
- Voter 2: Remove a $\rightarrow \{c\}$

Voter 3 in reduced game ($s_1 = a, s_2 = c$):

- $b=1, c=2 \rightarrow \{c\}$
- $\Rightarrow b$ is weakly dominated by c
- Voter 3: Remove b $\rightarrow \{c\}$

Thus the predicted outcome is votes: (a, c, c), which means that the candidate c wins via a majority.

Voter 1, despite having tie-breaking power, gets their least preferred outcome. Their power makes voting for 'a' undominated, but this predictability allows Voters 2 and 3 to coordinate around 'c' (V2's favorite, V3's second

favorite). This illustrates how apparent strength can become a strategic weakness.

